

Series

Engineering Mathematics–I (CSE-1001)
BTech 1st Year, IIIT Bhopal

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To the Student

Before you read this note, read the companion note on Sequences. This note continues directly from it. A series is nothing but a sequence in disguise. So, every tool from that note is used here, often on the very first page of a section. If the ϵ – N definition, the Monotone Convergence Theorem and the Ratio and Root Tests are not yet familiar, go back and settle them first.

This note explains **infinite series** in simple words. It is written for students who are meeting the topic properly for the first time. Every idea is built slowly. First we look at a real situation where the idea is needed, then we look at a picture, then we learn the definition, and only then do we work through examples. You do not need any advanced mathematics beyond the previous note and ordinary integration.

Read the sections in order. Everything in this chapter rests on one idea — that the sum of infinitely many numbers is *defined* as the limit of the partial sums, in Section 2. So do not rush past it. Keep pen and paper beside you: add the terms up yourself, and try each example on your own before reading its solution.

How to use these notes:

1. Read the companion note *Sequences* first, and keep it beside you — this note refers back to it often.
2. Attend the class lecture for each topic.
3. Read the corresponding section of these notes carefully, working through each example yourself.
4. Stop at every *Try it yourself* box and attempt it on the spot. There are seven of them, and each takes only two or three minutes.
5. Once you have finished a section, attempt the questions from the [companion Question Bank](#) for self-assessment.

Good luck, and enjoy the topic.

Notation. Throughout, $\mathbb{N} = \{1, 2, 3, \dots\}$ and $\mathbb{R}$ denotes the set of real numbers. We write $\sum a_n$ for $\sum_{n=1}^{\infty} a_n$ when the range is clear.

1 Why Study Series?

In the previous note we asked what happens to the *terms* of a sequence in the long run. Now we ask something harder: what happens when we *add them all up*?

Adding infinitely many numbers sounds impossible, and for a long time mathematicians thought it was. The surprise is that sometimes an infinite sum is a perfectly ordinary finite number, and sometimes it is not — and telling these two cases apart is one of the most useful skills in engineering mathematics.

Four situations, one piece of mathematics

1. A repeating decimal. What exactly *is* $0.3333\dots$? It is the infinite sum

$$\frac{3}{10} + \frac{3}{100} + \frac{3}{1000} + \dots$$

We all write it as $\frac{1}{3}$, but that equality needs a meaning, and giving it one is the business of this chapter.

2. How your calculator finds $\sin(0.5)$. There is no sine circuit inside it. It computes

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

and stops after a few terms. Two questions follow immediately: does this sum converge, and how many terms are enough for eight-digit accuracy?

3. A loan that never ends. A perpetual bond pays Rs. 1000 every year forever. Its value today is

$$\frac{1000}{1.08} + \frac{1000}{1.08^2} + \frac{1000}{1.08^3} + \dots$$

Infinitely many payments, yet a bank will quote a finite price. Why?

4. Signal processing. Any periodic signal — a musical note, a mains voltage, a square wave in a circuit — is written as a Fourier series, an infinite sum of sines and cosines. Every filter you design acts on that sum.

In each case the practical question — what is it worth, how many terms do I need, is the answer finite — becomes the mathematical question *does this series converge, and to what?* That is what this chapter answers.

The idea

Look hard at Figure 1. In both pictures the individual terms $\frac{1}{2^n}$ and $\frac{1}{n}$ tend to 0. Yet one total stops and the other does not. Whatever “the sum of infinitely many numbers” is going to mean, it cannot be decided by looking at the terms alone. It must be decided by looking at the *running totals* — and that is exactly how we shall define it.

2 From Sequences to Series: Partial Sums

We cannot add infinitely many numbers one after another; we would never finish. So we do the only honest thing available: we add finitely many, watch the running total, and

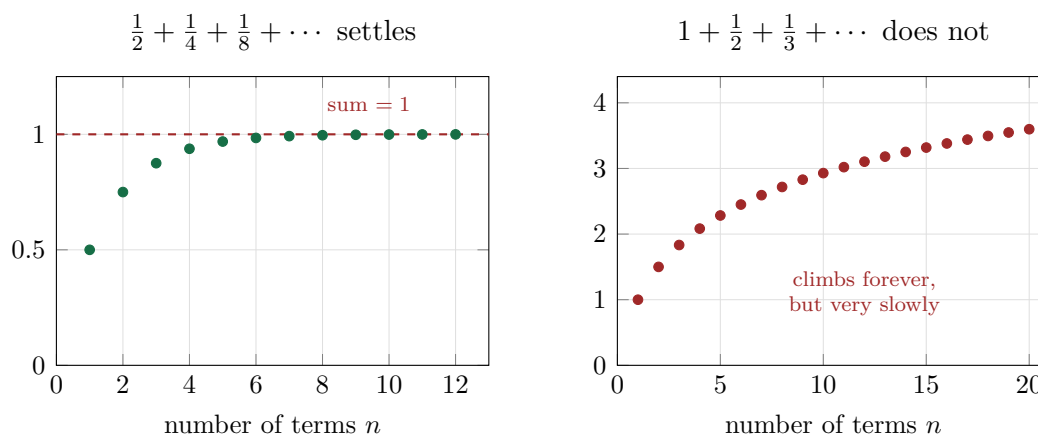


Figure 1: Two infinite sums whose terms both shrink to zero. On the left the running totals settle at 1; on the right they keep climbing without limit. The terms going to zero is clearly *not* enough to make a sum finite — that single observation is the reason this chapter needs so many tests.

take a limit.

Definition 2.1 (Series and partial sums). Let (a_n) be a sequence of real numbers. The n -th partial sum is

$$s_n = a_1 + a_2 + \cdots + a_n = \sum_{k=1}^n a_k.$$

The **infinite series** $\sum_{n=1}^{\infty} a_n$ is said to **converge** to S if the sequence of partial sums (s_n) converges to S , and we then write

$$\sum_{n=1}^{\infty} a_n = S = \lim_{n \rightarrow \infty} s_n.$$

If (s_n) diverges, the series is said to **diverge**.

A series is a sequence in disguise

This is the single most important sentence in the chapter. The symbol $\sum a_n$ looks like a new kind of object, but it is not: it is shorthand for the *sequence* $s_1, s_2, s_3, \dots$ of partial sums, and “the series converges” means nothing more and nothing less than “that sequence converges”.

Every theorem from the Sequences note is therefore available here at once. In particular, if the terms a_n are all positive then (s_n) is increasing, so by the Monotone Convergence Theorem the series converges *if and only if* the partial sums are bounded above. Almost every test in this chapter is that one fact in a different costume.

Example 2.2. For $a_n = \frac{1}{2^n}$ we have

$$s_n = \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = 1 - \frac{1}{2^n} \longrightarrow 1,$$

so $\sum_{n=1}^{\infty} \frac{1}{2^n} = 1$.

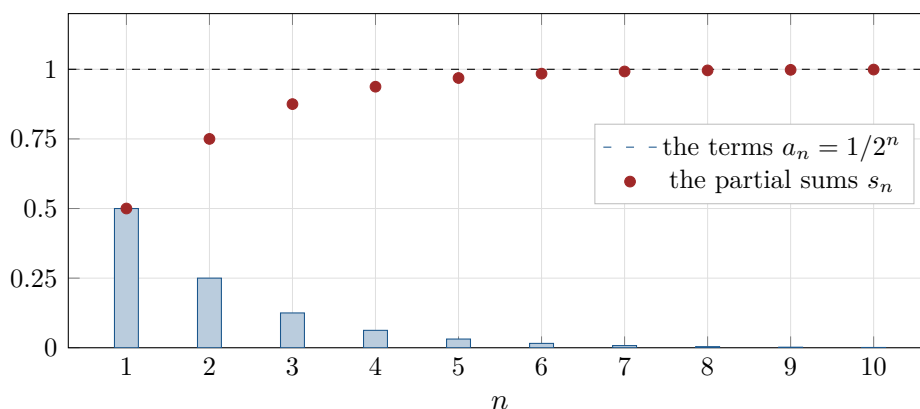


Figure 2: The bars are the terms; the dots are their running totals. Each dot sits exactly one bar-height above the previous dot. The bars shrink fast enough that the dots never pass the dashed line, so the series converges and its sum is 1.

Do not confuse a_n with s_n

The sequence of *terms* and the sequence of *partial sums* are different sequences, and they behave differently. In Figure 2, $a_n \rightarrow 0$ while $s_n \rightarrow 1$. The series $\sum a_n$ converges to 1, not to 0. Asking “does $\sum a_n$ converge?” is always a question about (s_n) , never about (a_n) — though, as we shall see in Section 5, the terms do carry one piece of useful information.

Try it yourself

Write down s_1, s_2, s_3 and s_4 for the series $\sum_{n=1}^{\infty} (-1)^{n+1} = 1 - 1 + 1 - 1 + \dots$. Does the sequence of partial sums converge? What does that tell you about the series?

3 The Geometric Series

Exactly one infinite series can be summed completely and easily, and almost every other result in this chapter is built by comparison with it. It is worth knowing perfectly.

Theorem 3.1 (Geometric series). *Let $a \neq 0$. The series*

$$\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + ar^3 + \dots$$

converges if and only if $|r| < 1$, and in that case

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$$

If $|r| \geq 1$ the series diverges.

Proof. For $r \neq 1$, multiply $s_n = a + ar + \dots + ar^{n-1}$ by r and subtract:

$$s_n - rs_n = a - ar^n \implies s_n = a \frac{1 - r^n}{1 - r}.$$

From the Sequences note, $r^n \rightarrow 0$ when $|r| < 1$, so $s_n \rightarrow \frac{a}{1-r}$. If $|r| > 1$ then $|r^n| \rightarrow \infty$ and (s_n) diverges; if $r = 1$ then $s_n = na \rightarrow \pm\infty$; and if $r = -1$ then s_n alternates between a and 0 , which does not converge. $\square$

Why $|r| < 1$ is exactly the right condition

Each term is a fixed fraction of the one before. If that fraction is smaller than 1 in size, the terms shrink geometrically — fast enough that the totals cannot run away. If it is 1 or larger, each term is at least as big as the last, and adding infinitely many such terms cannot settle. The borderline $|r| = 1$ fails, and it fails for two different reasons: at $r = 1$ the sum grows without limit, at $r = -1$ it oscillates.

Example 3.2. Evaluate $\sum_{n=1}^{\infty} \frac{3}{4^n}$.

Solution. Write out the first term: $n = 1$ gives $\frac{3}{4}$. Each later term is $\frac{1}{4}$ of the one before, so $a = \frac{3}{4}$ and $r = \frac{1}{4}$. Since $|r| < 1$,

$$\sum_{n=1}^{\infty} \frac{3}{4^n} = \frac{a}{1-r} = \frac{3/4}{1-1/4} = \frac{3/4}{3/4} = 1.$$

$\square$

Identify a correctly

The formula $\frac{a}{1-r}$ uses $a =$ *the first term that actually appears*, not the term at $n = 0$ unless the sum really starts there. For $\sum_{n=2}^{\infty} (1/3)^n$ the first term is $(1/3)^2 = \frac{1}{9}$, so the sum is $\frac{1/9}{1-1/3} = \frac{1}{6}$ — not $\frac{1}{1-1/3}$. When in doubt, write out the first two terms and divide to find r .

Three familiar things that are geometric series

The repeating decimal. $0.3333\dots$ has $a = \frac{3}{10}$ and $r = \frac{1}{10}$, so it equals $\frac{3/10}{1-1/10} = \frac{3/10}{9/10} = \frac{1}{3}$. The schoolbook fact is now a theorem. The same computation gives the

famous $0.9999\dots = \frac{9/10}{1-1/10} = 1$ — not “nearly 1”, but exactly 1.

The perpetual bond. Payments of P every year, discounted at rate i , are worth

$$\frac{P}{1+i} + \frac{P}{(1+i)^2} + \dots = \frac{P/(1+i)}{1-1/(1+i)} = \frac{P}{i}.$$

At $P = 1000$ and $i = 0.08$, the bond is worth 12,500 today. Infinitely many payments, a finite price — and a formula a banker can use.

The bouncing ball. From the Sequences note: a ball dropped from 2 m rebounds to 60% each time. The total distance travelled is

$$2 + 2(2(0.6) + 2(0.6)^2 + \dots) = 2 + 4 \cdot \frac{0.6}{1-0.6} = 2 + 6 = 8 \text{ m.}$$

Infinitely many bounces, eight metres of travel, and a finite time.

Try it yourself

Evaluate $\sum_{n=0}^{\infty} \frac{(-1)^n}{5^n}$ and $\sum_{n=3}^{\infty} \left(\frac{2}{3}\right)^n$.

Answers: $\frac{5}{6}$ and $\frac{8}{9}$.

4 Telescoping Series

The second family we can sum exactly is the one where almost everything cancels.

Example 4.1. Show that $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$.

Solution. Split the term by partial fractions:

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Now write out the partial sum and watch the middle collapse:

$$s_n = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{n+1}.$$

Hence $s_n \rightarrow 1$, and the series converges to 1. $\square$

Why this matters more than it looks

Telescoping is the only other situation in which we get the partial sum s_n in closed form, and therefore the exact value of the sum. Every remaining test in this chapter answers only the yes/no question “does it converge?” and tells us nothing about the value. Knowing a sum converges and knowing what it equals are two quite different achievements — and the second is usually much harder.

5 The n -th Term Test

Before applying any real test, there is one cheap check that sometimes finishes the problem in a single line.

Theorem 5.1 (n -th term test for divergence). *If $\sum a_n$ converges, then $a_n \rightarrow 0$. Equivalently, in the form we actually use:*

$$\text{if } a_n \not\rightarrow 0, \text{ then } \sum a_n \text{ diverges.}$$

Proof. Suppose $s_n \rightarrow S$. Then $s_{n-1} \rightarrow S$ as well, and $a_n = s_n - s_{n-1} \rightarrow S - S = 0$. $\square$

Example 5.2. $\sum_{n=1}^{\infty} \frac{n}{2n+1}$ diverges, because $\frac{n}{2n+1} \rightarrow \frac{1}{2} \neq 0$. No further work is needed.

Example 5.3. $\sum_{n=1}^{\infty} (-1)^n$ diverges, because $(-1)^n$ does not converge to anything, let alone to 0.

The converse is false — and this is the classic trap

Theorem 5.1 runs in one direction only. From $a_n \rightarrow 0$ you may conclude **nothing at all**. The harmonic series $\sum \frac{1}{n}$ has $\frac{1}{n} \rightarrow 0$ and yet diverges (Section 6); the series $\sum \frac{1}{n^2}$ also has terms tending to 0 and converges. Both cases occur, so the test is silent.

Written as a slogan: $a_n \rightarrow 0$ is *necessary for convergence, but not sufficient*. Every year students write “the terms tend to zero, so the series converges”. It earns no marks, because it is not a valid argument.

Try it yourself

Which of these can be settled immediately by the n -th term test, and what is the verdict in those cases?

$$(i) \sum \frac{3n^2 + 1}{n^2 + 5} \quad (ii) \sum \frac{1}{\sqrt{n}} \quad (iii) \sum \cos\left(\frac{1}{n}\right) \quad (iv) \sum \frac{n!}{2^n}$$

6 The Harmonic Series

This one series deserves a section of its own. It is the standard counterexample in the whole subject, and the argument that settles it is a small masterpiece.

Theorem 6.1 (Divergence of the harmonic series). *The series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.*

Proof (Oresme, around 1350). Group the terms in blocks whose lengths double:

$$1 + \underbrace{\frac{1}{2}}_{\geq 1/2} + \underbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}_{\geq 2 \cdot \frac{1}{4} = \frac{1}{2}} + \underbrace{\left(\frac{1}{5} + \cdots + \frac{1}{8}\right)}_{\geq 4 \cdot \frac{1}{8} = \frac{1}{2}} + \underbrace{\left(\frac{1}{9} + \cdots + \frac{1}{16}\right)}_{\geq 8 \cdot \frac{1}{16} = \frac{1}{2}} + \cdots$$

Each bracket contains terms no smaller than its last one, and the number of terms exactly compensates, so every bracket is at least $\frac{1}{2}$. Therefore

$$s_{2^k} \geq 1 + \frac{k}{2},$$

which is unbounded. An increasing unbounded sequence of partial sums diverges to $+\infty$. $\square$

How slowly is “slowly”?

Because $H_n \approx \ln n + \gamma$, reaching a total of 10 needs about 12,367 terms; reaching 100 needs more terms than there are atoms in a human body. A computer adding a million terms per second would take longer than the age of the universe to push the total past 50.

And yet it is unbounded, with a consequence you can test on a table. Stack identical blocks at the edge of a desk, each shifted forward. The furthest the top block can overhang, using n blocks, is $\frac{1}{2}H_n$ block-lengths. Since H_n is unbounded, **a tall enough stack can overhang by any distance you like** — a whole block, ten

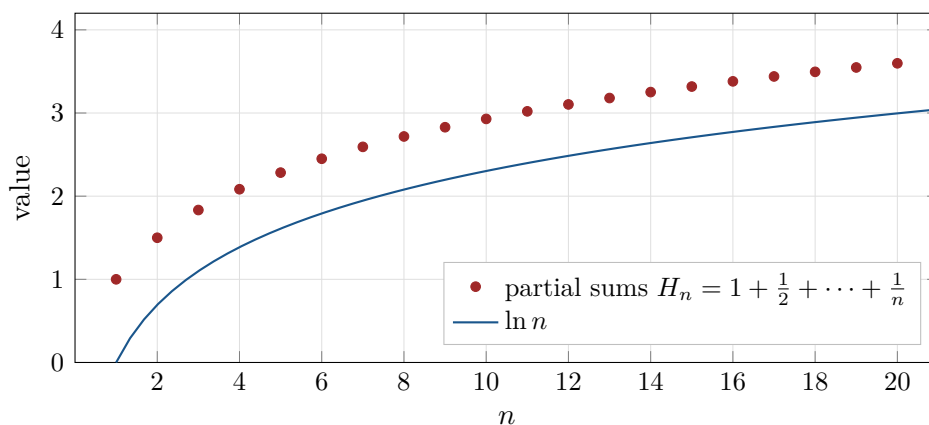


Figure 3: The harmonic partial sums grow like $\ln n$ — they do reach every height eventually, but at a crawl. The gap between the two curves settles at Euler’s constant $\gamma \approx 0.5772$.

blocks, the width of the room. You would need about 272 million blocks to overhang by ten block-lengths, which is why nobody has seen it done.

The lesson to carry forward

The harmonic series sits exactly on the boundary between convergence and divergence. Terms slightly smaller — $\frac{1}{n^{1.01}}$ — give a convergent series; terms slightly larger — $\frac{1}{n^{0.99}}$ — give a divergent one. This is why $\sum \frac{1}{n}$ turns up as the comparison case in so many problems, and why the p -series test of Section 8 is worth memorising.

7 Comparison Tests

From here on, all series have **non-negative terms** unless stated otherwise. For such a series the partial sums increase, so by the Monotone Convergence Theorem the only question is whether they are bounded — and the easiest way to bound them is to compare with a series we already understand.

Theorem 7.1 (Direct comparison test). *Suppose $0 \leq a_n \leq b_n$ for all $n \geq N_0$. Then*

- (i) *if $\sum b_n$ converges, so does $\sum a_n$;*
- (ii) *if $\sum a_n$ diverges, so does $\sum b_n$.*

The idea

Read the two parts as one sentence: *smaller than convergent is convergent; bigger than divergent is divergent*. The other two combinations tell you nothing — being bigger than a convergent series, or smaller than a divergent one, is no information at all.

Example 7.2. Show that $\sum_{n=1}^{\infty} \frac{1}{n^2 + 3n}$ converges.

Solution. For all $n \geq 1$ we have $n^2 + 3n > n^2 > 0$, so

$$0 < \frac{1}{n^2 + 3n} < \frac{1}{n^2}.$$

We shall see in Section 8 that $\sum \frac{1}{n^2}$ converges, so by comparison our series converges too. $\square$

Example 7.3. Show that $\sum_{n=2}^{\infty} \frac{1}{\ln n}$ diverges.

Solution. For $n \geq 2$ we have $\ln n < n$, so $\frac{1}{\ln n} > \frac{1}{n} > 0$. Since $\sum \frac{1}{n}$ diverges, so does the larger series. $\square$

Direct comparison needs a clean inequality, which is not always available. The next test removes that difficulty by comparing growth rates instead.

Theorem 7.4 (Limit comparison test). *Let $a_n > 0$ and $b_n > 0$, and suppose*

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c, \quad 0 < c < \infty.$$

Then $\sum a_n$ and $\sum b_n$ either both converge or both diverge.

How to choose b_n

Keep only the dominant power in the numerator and the denominator of a_n and throw everything else away; the result is your b_n . For $a_n = \frac{2n^2 - n + 3}{5n^4 + n^2}$, the dominant behaviour is $\frac{2n^2}{5n^4} \sim \frac{1}{n^2}$, so take $b_n = \frac{1}{n^2}$. The limit comparison test then does the rest without any inequality juggling.

Example 7.5. Test $\sum_{n=1}^{\infty} \frac{3n+2}{n^3+5n+1}$.

Solution. The dominant behaviour is $\frac{3n}{n^3} = \frac{3}{n^2}$, so take $b_n = \frac{1}{n^2}$. Then

$$\frac{a_n}{b_n} = \frac{(3n+2)n^2}{n^3+5n+1} = \frac{3n^3+2n^2}{n^3+5n+1} = \frac{3+\frac{2}{n}}{1+\frac{5}{n^2}+\frac{1}{n^3}} \rightarrow 3,$$

which is finite and non-zero. Since $\sum \frac{1}{n^2}$ converges, so does the given series. $\square$

Example 7.6. Test $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2+1}}$.

Solution. Take $b_n = \frac{1}{n}$. Then $\frac{a_n}{b_n} = \frac{n}{\sqrt{n^2+1}} \rightarrow 1$, finite and non-zero. Since $\sum \frac{1}{n}$ diverges, the given series diverges. $\square$

Comparison tests need non-negative terms

Both tests above assume the terms are non-negative (from some point onwards). They are simply false for series that change sign: $\sum \frac{(-1)^n}{n}$ converges while $\sum \frac{1}{n}$ diverges, even though the terms have the same size. Series with mixed signs are handled in Sections 10 and 11.

8 The Integral Test and p -Series

The comparison tests need something to compare with. The integral test manufactures the comparison out of an area, and it is what finally settles $\sum 1/n^p$ for every p .

Theorem 8.1 (Integral test). *Let f be positive, continuous and decreasing on $[1, \infty)$, and put $a_n = f(n)$. Then*

$$\sum_{n=1}^{\infty} a_n \quad \text{and} \quad \int_1^{\infty} f(x) dx$$

either both converge or both diverge.

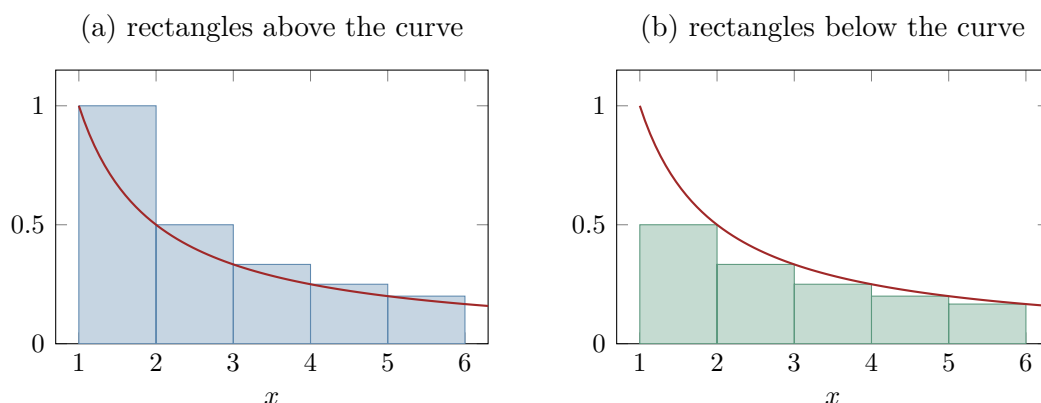


Figure 4: The integral test for $f(x) = 1/x$. In (a) each rectangle has height $f(k)$ and width 1, so the shaded area is $a_1 + \dots + a_5$ and it *covers* the area under the curve. In (b) the rectangles have height $f(k+1)$ and lie *under* the curve. Hence $\int_1^{n+1} f \leq s_n \leq a_1 + \int_1^n f$, and the sum and the integral must live or die together.

Why a picture is a proof here

Figure 4 contains the entire argument. Picture (a) says the partial sums are at least as big as the integral, so if the integral is infinite the sums are too. Picture (b) says the partial sums are at most a_1 plus the integral, so if the integral is finite the sums are bounded — and bounded increasing partial sums converge, by the Monotone Convergence Theorem. That is the whole proof.

Theorem 8.2 (p -series test). *For $p > 0$,*

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \quad \text{converges if } p > 1, \quad \text{diverges if } p \leq 1.$$

Proof. Apply the integral test to $f(x) = x^{-p}$, which is positive, continuous and decreasing. For $p \neq 1$,

$$\int_1^b x^{-p} dx = \frac{b^{1-p} - 1}{1-p},$$

which tends to $\frac{1}{p-1}$ when $p > 1$ (finite) and to $+\infty$ when $p < 1$. For $p = 1$, $\int_1^b \frac{dx}{x} = \ln b \rightarrow \infty$. $\square$

Example 8.3. $\sum \frac{1}{n^2}$ converges ($p = 2 > 1$); $\sum \frac{1}{\sqrt{n}}$ diverges ($p = \frac{1}{2} \leq 1$); $\sum \frac{1}{n}$ diverges ($p = 1$); $\sum \frac{1}{n^{1.001}}$ converges.

The Basel problem

The integral test tells us $\sum \frac{1}{n^2}$ converges, but not to what. Finding the value defeated the best mathematicians of the seventeenth century and became known as the Basel problem. Euler settled it in 1735:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \approx 1.6449.$$

That π should appear in a sum of reciprocal squares, with no circle anywhere in sight, is one of the genuine surprises of mathematics. It is also a reminder of the point made in Section 4: proving a series converges is routine; finding its sum can take a century.

Try it yourself

Use the integral test to decide whether $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ converges.

Hint: substitute $u = \ln x$. Answer: it diverges — just.

9 Ratio and Root Tests

These are the same two tests met in the Sequences note, but they are now asked a different question, and it is important to see what has changed.

Theorem 9.1 (Ratio test for series). *Let $a_n \neq 0$ and suppose $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$ exists. Then*

- (i) $L < 1 \implies \sum a_n$ converges absolutely;
- (ii) $L > 1 \implies \sum a_n$ diverges;
- (iii) $L = 1 \implies$ the test gives no information.

Theorem 9.2 (Root test for series). *Suppose $\lim_{n \rightarrow \infty} |a_n|^{1/n} = L$ exists. Then the same three conclusions hold.*

Same test, different question

In the Sequences note, $L < 1$ told us that the *terms* a_n tend to 0. Here, $L < 1$ tells us the much stronger fact that the *sum* is finite. The reason is the geometric series:

if the ratio settles below 1, then from some point onwards $|a_n|$ is dominated by a convergent geometric series, and direct comparison finishes the job. Everything in this section is Theorem 3.1 wearing a disguise.

Example 9.3. Test $\sum_{n=1}^{\infty} \frac{2^n}{n!}$.

Solution.

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} = \frac{2}{n+1} \rightarrow 0 < 1,$$

so the series converges. (Its sum, in fact, is $e^2 - 1$.) □

Example 9.4. Test $\sum_{n=1}^{\infty} \frac{n!}{10^n}$.

Solution.

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)!}{10^{n+1}} \cdot \frac{10^n}{n!} = \frac{n+1}{10} \rightarrow \infty > 1,$$

so the series diverges. Notice that the terms themselves grow without bound, so the n -th term test would also have settled it. □

Example 9.5. Test $\sum_{n=1}^{\infty} \left(\frac{3n+1}{4n+5} \right)^n$.

Solution. The whole expression is an n -th power, so use the root test:

$$|a_n|^{1/n} = \frac{3n+1}{4n+5} = \frac{3 + \frac{1}{n}}{4 + \frac{5}{n}} \rightarrow \frac{3}{4} < 1,$$

so the series converges. □

L = 1 means “try something else”

At $L = 1$ the ratio test has failed, which is not the same as the series having failed. Both $\sum \frac{1}{n}$ (divergent) and $\sum \frac{1}{n^2}$ (convergent) give $L = 1$. Writing “ $L = 1$, hence divergent” is wrong. When $L = 1$ — which happens for *every* p -series and every rational function of n — abandon the ratio test and use comparison, limit comparison, or the integral test instead.

Try it yourself

Decide the behaviour of each, choosing your test deliberately.

$$(i) \sum \frac{n^3}{3^n} \quad (ii) \sum \frac{n^2 + 1}{n^4 + 3} \quad (iii) \sum \frac{(n!)^2}{(2n)!} \quad (iv) \sum \frac{1}{n^{2/3}}$$

10 Alternating Series

All the tests so far demanded non-negative terms. We now allow the sign to flip at every step, which turns out to make convergence *easier*, not harder.

Definition 10.1 (Alternating series). A series of the form

$$\sum_{n=1}^{\infty} (-1)^{n+1} b_n = b_1 - b_2 + b_3 - b_4 + \cdots, \quad b_n > 0,$$

is called an **alternating series**.

Theorem 10.2 (Leibniz test). *If the sequence (b_n) satisfies*

(i) b_n is decreasing, i.e. $b_{n+1} \leq b_n$ for all n (or from some point onwards), and

(ii) $b_n \rightarrow 0$,

then $\sum (-1)^{n+1} b_n$ converges. Moreover, if S is its sum then

$$|S - s_n| \leq b_{n+1},$$

so the error after n terms is no larger than the first term omitted.

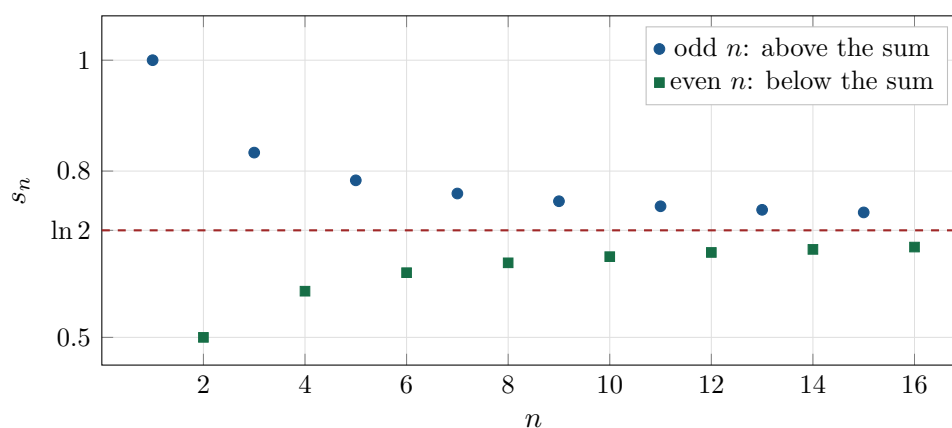


Figure 5: Partial sums of the alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \cdots$. Each step overshoots the limit and each overshoot is smaller than the last, so the sum is trapped between consecutive partial sums. That trapping is exactly the error bound $|S - s_n| \leq b_{n+1}$.

Why alternating helps

With all-positive terms, every new term pushes the total further up and the only hope is that the pushes shrink fast enough. With alternating terms, each push is in the opposite direction to the last, so the partial sums straddle the answer and close in on it like a pair of pincers (Figure 5). Shrinking to zero is then enough on its own, however slowly it happens.

Example 10.3 (The alternating harmonic series). The series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ converges, since $b_n = \frac{1}{n}$ decreases to 0. Its sum is $\ln 2 \approx 0.6931$.

This is worth pausing on. Remove the signs and the series diverges to infinity; keep them and it converges to $\ln 2$. The signs are doing all the work.

The error bound is what engineers actually use

Theorem 10.2 does something the other tests never do: it tells you *how wrong* you are if you stop early. To compute $\ln 2$ to three decimal places from the alternating harmonic series you need $b_{n+1} = \frac{1}{n+1} < 0.0005$, so $n > 1999$ — two thousand terms for three decimals, which is hopeless. By contrast the series for $\sin x$ has $b_n = \frac{x^{2n-1}}{(2n-1)!}$, and at $x = 0.5$ the fourth term is already below 10^{-8} . This is why calculators use Taylor series and not this one: the error bound decides the design.

Both conditions must be checked

The Leibniz test needs b_n decreasing *and* $b_n \rightarrow 0$. Dropping either one breaks it. For $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} - \frac{1}{2} + \dots$ the terms tend to 0 along the odd places but the sequence b_n is not decreasing, and the series diverges. And if $b_n \not\rightarrow 0$, the n -th term test already guarantees divergence, whatever the signs do.

Try it yourself

How many terms of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ are needed to be sure of the sum to within 0.01?

Answer: $n = 10$, since $b_{11} = 1/121 < 0.01$.

11 Absolute and Conditional Convergence

The alternating harmonic series converges, but only because of its signs. That distinction is important enough to have a name.

Definition 11.1 (Absolute and conditional convergence). The series $\sum a_n$ **converges absolutely** if $\sum |a_n|$ converges. It **converges conditionally** if $\sum a_n$ converges but $\sum |a_n|$ diverges.

Theorem 11.2 (Absolute convergence implies convergence). If $\sum |a_n|$ converges, then $\sum a_n$ converges.

Why the distinction earns its keep

Absolute convergence is the robust kind: the series converges because the terms are genuinely small, and no amount of sign-shuffling can disturb it. Conditional convergence is fragile: the series converges only because of cancellation between positive and negative parts, and that cancellation is easy to destroy.

The practical consequence is that $\sum |a_n|$ can be attacked with every test in Sections 7–9, all of which need non-negative terms. So the standard strategy for a series with mixed signs is: first test $\sum |a_n|$; if that converges you are finished, and only if it diverges do you need the Leibniz test.

Example 11.3. $\sum \frac{(-1)^{n+1}}{n^2}$ converges absolutely, because $\sum \frac{1}{n^2}$ converges by the p -series test.

Example 11.4. $\sum \frac{(-1)^{n+1}}{n}$ converges conditionally: it converges by the Leibniz test, but $\sum \frac{1}{n}$ diverges.

Rearrangement: a genuinely alarming theorem

Finite sums may be added in any order. Conditionally convergent series may not. Riemann proved in 1853 that **a conditionally convergent series can be rearranged to sum to any number whatsoever** — to 7, to $-\pi$, to $+\infty$.

Concretely, the alternating harmonic series adds to $\ln 2 \approx 0.693$ in its usual order; take its terms in a different order — two positives, then a negative, repeatedly — and the same numbers add to $\frac{3}{2} \ln 2$. Nothing was added or removed; only the order changed.

Absolutely convergent series are immune: any rearrangement of one gives the same sum. So the distinction is not academic book-keeping. It decides whether the phrase “the sum” even refers to one definite number.

12 Power Series

We close with the payoff. Everything so far has been about summing numbers; now we sum functions, and the result is how a computer evaluates almost every function it knows.

Definition 12.1 (Power series). A **power series** centred at a is a series of the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots,$$

where the c_n are constants. For each fixed x this is an ordinary series of numbers, which may converge or diverge.

Theorem 12.2 (Radius of convergence). *For any power series there is an R with $0 \leq R \leq \infty$, called the **radius of convergence**, such that the series converges absolutely for $|x-a| < R$ and diverges for $|x-a| > R$. When the limit exists,*

$$R = \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|.$$

At the two endpoints $|x-a| = R$ the series may do either, and each must be checked separately.

Example 12.3. Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$.

Solution. By the ratio test,

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{x^{n+1}}{n+1} \cdot \frac{n}{x^n} \right| = |x| \frac{n}{n+1} \rightarrow |x|,$$

so the series converges for $|x| < 1$ and diverges for $|x| > 1$: hence $R = 1$. Now check the endpoints.

- At $x = 1$: the series is $\sum \frac{1}{n}$, which diverges.
- At $x = -1$: the series is $\sum \frac{(-1)^n}{n}$, which converges by the Leibniz test.

The interval of convergence is therefore $[-1, 1)$. □

Never forget the endpoints

The ratio test is silent at $|x - a| = R$, because there $L = 1$. The two endpoints must be substituted in and tested individually, and they genuinely can differ — as the example above shows, where one endpoint converges and the other does not. An answer of “ $-1 < x < 1$ ” when the true interval is $[-1, 1)$ loses marks, and the marks are in the endpoints.

12.1 Taylor and Maclaurin series

The most useful power series are those that reproduce a known function. If f has derivatives of all orders at a , its **Taylor series** is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n,$$

and when $a = 0$ this is called the **Maclaurin series**. The five worth memorising are:

Series	Valid for
$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \cdots$	all x
$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$	all x
$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$	all x
$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$	$ x < 1$
$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$	$-1 < x \leq 1$

How your calculator really finds $\sin(0.5)$

It adds up the Maclaurin series and stops when the next term is too small to matter:

$$0.5 - \frac{0.5^3}{6} + \frac{0.5^5}{120} - \frac{0.5^7}{5040} = 0.5 - 0.0208333 + 0.0002604 - 0.0000016 = 0.4794255.$$

The true value is $0.4794255386\dots$, so four terms give seven correct decimals. The alternating series error bound of Section 10 is what guarantees this in advance — the machine knows how many terms it needs *before* it computes them.

Notice the last row of the table: setting $x = 1$ in the series for $\ln(1+x)$ gives $\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \cdots$, which is exactly the alternating harmonic series. The $\ln 2$ that appeared in Section 10 was never a coincidence.

Try it yourself

Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{(x-2)^n}{3^n}$ and state the interval, endpoints in-

cluded.

Answer: $R = 3$, interval $(-1, 5)$; both endpoints diverge.

13 Choosing a Test: A Strategy

Students rarely lose marks because they cannot apply a test. They lose marks because they apply the wrong one. Work down this list in order and stop at the first test that fits.

1. **Does $a_n \rightarrow 0$?** If not, the series diverges by the n -th term test, and you are finished in one line. If it does, this tells you nothing — carry on.
2. **Is it geometric or a p -series?** Then quote Theorem 3.1 or Theorem 8.2 and stop. Between them these cover a surprising share of examination questions.
3. **Does the term telescope?** If partial fractions split a_n into a difference, compute s_n directly and take the limit — this gives the sum, not just convergence.
4. **Are there factorials, or is the whole term an n -th power?** Use the ratio test for factorials, the root test for n -th powers.
5. **Is a_n a rational function of n , or close to one?** Ratio and root will both give $L = 1$ and fail. Use limit comparison against the dominant power.
6. **Does the sign alternate?** Test $\sum |a_n|$ first; if it converges you have absolute convergence. If not, apply the Leibniz test for conditional convergence.
7. **Otherwise**, if $a_n = f(n)$ for a positive decreasing f you can integrate, use the integral test.

Convergence is not the value

Tests 4 to 7 answer only “does it converge?”. They never produce the sum. Only geometric series, telescoping series and the standard Taylor series of Section 12 give an actual value. A question asking you to *evaluate* a series is therefore asking for one of those three, not for a test.

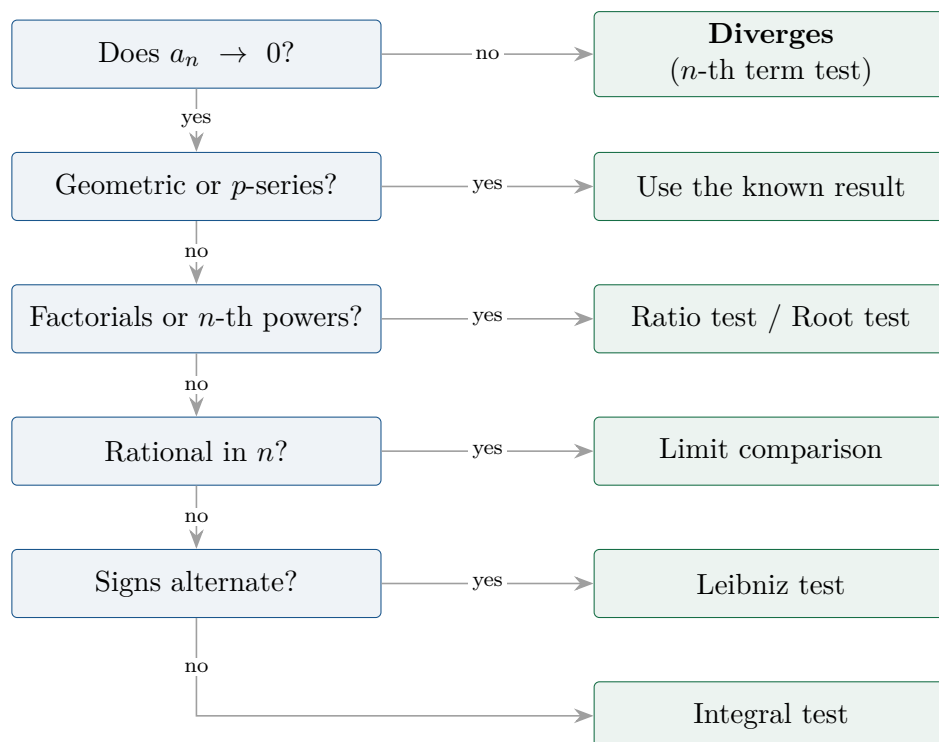


Figure 6: The same strategy as a decision chart. Enter at the top and leave at the first branch that applies.

14 Standard Series Worth Knowing

Series	Behaviour	Sum, if known
$\sum_{n=0}^{\infty} ar^n$	converges iff $ r < 1$	$\frac{a}{1-r}$
$\sum 1/n^p$	converges iff $p > 1$	—
$\sum 1/n$	diverges	—
$\sum 1/n^2$	converges	$\pi^2/6$
$\sum 1/(n \ln n)$	diverges	—
$\sum \frac{1}{n(n+1)}$	converges (telescoping)	1
$\sum \frac{(-1)^{n+1}}{n}$	converges conditionally	$\ln 2$
$\sum \frac{1}{n!}$	converges	$e - 1$
$\sum \frac{x^n}{n!}$	converges for all x	e^x
$\sum \frac{a^n}{n!}$	converges for all a	$e^a - 1$ (from $n = 1$)

Two boundary cases to remember

The pair $\sum \frac{1}{n}$ (diverges) and $\sum \frac{1}{n^2}$ (converges) marks the frontier, and $\sum \frac{1}{n \ln n}$ shows how fine the frontier is: adding even the slowly growing factor $\ln n$ to the denominator of $\frac{1}{n}$ is still not enough to bring the sum under control. Anything that behaves like $\frac{1}{n}$ or worse diverges; anything that behaves like $\frac{1}{n^{1+\delta}}$ for some $\delta > 0$ converges.

15 Summary

The logical structure of the chapter is shown in Figure 7; the statements are collected below it.

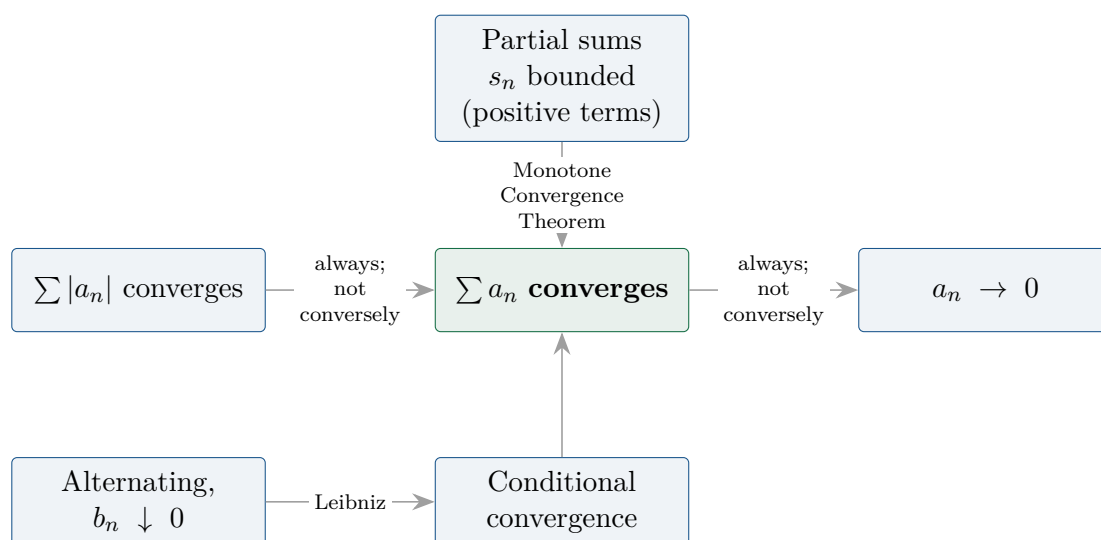


Figure 7: How the main results fit together. Convergence sits at the centre: bounded partial sums and absolute convergence are each sufficient for it, $a_n \rightarrow 0$ is necessary but never sufficient, and the Leibniz test supplies the conditional case that absolute convergence misses.

- A **series** $\sum a_n$ is defined to be the limit of its sequence of **partial sums** $s_n = a_1 + \cdots + a_n$. Every question about a series is a question about that sequence.
- For **positive terms**, (s_n) increases, so the series converges if and only if the partial sums are bounded. This is the Monotone Convergence Theorem, and it underlies every test below.
- A **geometric series** $\sum ar^n$ converges iff $|r| < 1$, with sum $\frac{a}{1-r}$. A **telescoping series** is summed by collapsing s_n . These are the only two elementary families whose sums we can evaluate.
- The **n -th term test**: if $a_n \not\rightarrow 0$ the series diverges. The converse is false — $\sum \frac{1}{n}$ is the standard counterexample.
- The **harmonic series** diverges, but like $\ln n$; it marks the boundary between convergence and divergence.

- **Comparison** and **limit comparison** settle series with non-negative terms by measuring them against a known one, usually a p -series.
- The **integral test** gives the p -series rule: $\sum 1/n^p$ converges iff $p > 1$.
- The **ratio** and **root** tests give absolute convergence when $L < 1$ and divergence when $L > 1$; both are useless at $L = 1$, which is exactly the rational-function case.
- The **Leibniz test**: an alternating series with b_n decreasing to 0 converges, and the error after n terms is at most b_{n+1} . This is the only test that also bounds the error.
- **Absolute convergence** implies convergence and is stable under rearrangement; **conditional convergence** is not, and by Riemann's theorem its terms can be reordered to reach any sum at all.
- A **power series** converges on an interval of radius R about its centre, with the end-points always to be checked separately. The Taylor series of e^x , $\sin x$, $\cos x$, $\frac{1}{1-x}$ and $\ln(1+x)$ are how functions are actually computed.

Note: After completing these notes, proceed to the [companion Question Bank](#) for self-assessment.

Best wishes.
